

Matter-Dependent Vacuum Energy Density and Inhomogeneous Cosmic Expansion

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Abstract

We propose that the vacuum energy density of spacetime is a decreasing function of the local matter density: $\Lambda(\rho_m) = \Lambda_0 - \alpha \rho_m$, where Λ_0 is the energy density of matter-free vacuum and α is a dimensionless coupling parameter. This ansatz is motivated by three established mechanisms through which real matter suppresses vacuum fluctuations: Pauli exclusion of fermionic modes, vacuum polarization by charged matter, and modification of the fluctuation spectrum in curved spacetime. The resulting model predicts inhomogeneous cosmic expansion—faster in voids (where vacuum energy is unsuppressed) and slower in walls and filaments (where matter suppresses it)—producing the observed cosmic web through a self-reinforcing feedback loop. We derive modified Friedmann and Poisson equations, construct a two-component (void + wall) toy model, and extract four testable predictions: (1) enhanced void expansion profiles measurable through peculiar velocities, (2) amplified integrated Sachs–Wolfe effect from voids, (3) a natural account of the S_8 tension between early- and late-universe measurements, and (4) accelerated early structure formation consistent with JWST observations of massive high-redshift galaxies. The model reinterprets the supernova-based evidence of Seifert et al. (2025) for differential expansion as support for inhomogeneous vacuum energy rather than for the complete elimination of dark energy proposed by timescape cosmology. The entire framework is governed by a single new parameter α , which can be independently constrained by each of the four observational tests. Constraints from Big Bang nucleosynthesis, matter density evolution, and the running vacuum model literature narrow the viable range to $\alpha \sim 0.01$ – 0.05 . We show that the cosmic web constitutes a self-consistent fixed-point configuration of the cyclic operator $\rho_m \rightarrow \Lambda \rightarrow g_{\mu\nu} \rightarrow \rho_m$, whose existence is guaranteed under standard continuity and compactness conditions.

Keywords: vacuum energy, dark energy, inhomogeneous expansion, cosmic web, large-scale structure, cosmological constant, backreaction

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1 Introduction

The standard Λ CDM cosmological model describes the universe as a homogeneous, isotropic spacetime whose late-time expansion is driven by a cosmological constant Λ with a fixed energy density of approximately 68% of the total energy budget (Planck Collaboration, 2020). This framework has proven remarkably successful in accounting for a wide range of observations, from the cosmic microwave background (CMB) anisotropies to baryon acoustic oscillations (BAO) and the distance–redshift relation of Type Ia supernovae (Riess et al., 1998; Perlmutter et al., 1999).

However, several persistent tensions have emerged. The locally measured Hubble constant $H_0 \approx 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Riess et al., 2022) disagrees with the CMB-inferred value $H_0 \approx 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Planck Collaboration, 2020) at the 5σ level. The amplitude of matter fluctuations at late times, parameterized by $S_8 = \sigma_8(\Omega_m/0.3)^{0.5}$, measured through weak gravitational lensing, is systematically lower than the CMB prediction (Di Valentino et al., 2021). Observations by the James Webb Space Telescope (JWST) have revealed unexpectedly massive galaxies at redshifts $z > 10$, challenging the timeline of structure formation predicted by Λ CDM (Labbé et al., 2023). The Dark Energy Spectroscopic Instrument (DESI) has reported hints that the dark energy equation of state w deviates from -1 (DESI Collaboration, 2024).

Independently, Seifert et al. (2025) demonstrated that the Pantheon+ supernova data prefer a model with differential expansion—faster in voids, slower in dense regions—over the standard homogeneous Λ CDM. Their timescape cosmology (Wiltshire, 2007) interprets this as evidence that dark energy does not exist and that the apparent acceleration is an artifact of averaging inhomogeneous expansion. The present author has critically evaluated this claim (Kriger, 2025a), arguing that the timescape interpretation is quantitatively insufficient: CMB and BAO data independently require a real dark energy component, which timescape cannot accommodate.

The present work proposes a middle path. We accept the observational evidence for differential expansion but retain dark energy as a real physical phenomenon—the energy of the quantum vacuum. Our central hypothesis is that the vacuum energy density is not spatially uniform, but depends on the local matter density:

$$\Lambda(\rho_m) = \Lambda_0 - \alpha \rho_m, \quad (1)$$

where Λ_0 is the vacuum energy density in the absence of matter, and $\alpha > 0$ is a dimensionless coupling parameter governing the strength of suppression. (Both Λ_0 and ρ_m carry dimensions of energy density, [energy/volume]; their difference is an energy density, so α is indeed dimensionless. Throughout this paper, we work in units where $8\pi G = c = 1$ unless otherwise stated, so that all densities are measured in the same units as the Friedmann equation parameter H^2 .)

This is not a new dynamical field. It is a consequence of established physics: real matter modifies the structure of the quantum vacuum, suppressing its fluctuation modes and thereby reducing its energy density. Three independent mechanisms contribute to this suppression, and all three operate in the same direction. The resulting model produces inhomogeneous expansion from a single parameter α , preserves consistency with CMB and BAO data (which

probe the early, nearly homogeneous universe), and generates testable predictions that differ from both Λ CDM and timescape cosmology.

The theoretical underpinning draws on a broader structural program. The necessity of a dynamic metric—one that depends on the distribution of state density—was established from general conditions on sustainable complexity in Kriger (2025b). The mathematical existence of self-consistent cyclic configurations (matter $\rightarrow$ vacuum $\rightarrow$ metric $\rightarrow$ matter) was proven via fixed-point theorems in Kriger (2025c). The impossibility of a state with zero vacuum energy—equivalent to the impossibility of absolute nothingness—was demonstrated through four independent formal routes in Kriger (2025d). The present paper provides the specific physical realization of these structural results in the cosmological context. (We note that these three works are available on Zenodo as preprints and have not undergone traditional peer review. The present paper is self-contained and does not depend on their results for any of its quantitative claims; the references provide additional theoretical context.)

The paper is organized as follows. Section 2 presents the physical motivation, detailing the three mechanisms of vacuum suppression. Section 3 develops the mathematical framework. Section 4 derives four observational predictions. Section 5 connects the model to existing results, including Seifert et al. (2025) and DESI. Section 6 presents toy models with quantitative illustrations. Section 7 discusses implications and limitations. Section 8 summarizes the results.

2 Physical Motivation

The quantum vacuum is not empty space. In quantum field theory (QFT), the vacuum is the lowest-energy state of all quantum fields, but this state is dynamically rich: it supports zero-point fluctuations of every field mode, virtual particle–antiparticle pair creation, and vacuum polarization effects. These processes are not hypothetical—they produce measurable consequences, including the Lamb shift in hydrogen, the anomalous magnetic moment of the electron, and the Casimir force between conducting plates.

The energy associated with these fluctuations—the vacuum energy density—is the quantity identified with the cosmological constant Λ in general relativity. In Λ CDM, this quantity is assumed to be spatially uniform: the same in a void as in a galaxy cluster, the same near a black hole as in intergalactic space. We argue that this assumption is physically unjustified. Three independent, well-established mechanisms ensure that the presence of real matter modifies the vacuum state and reduces its energy density.

2.1 Mechanism I: Pauli Exclusion of Fermionic Modes

The vacuum fluctuation spectrum includes virtual fermion–antifermion pairs (electron–positron, quark–antiquark) that momentarily appear and annihilate. Each such virtual pair occupies quantum states characterized by position, momentum, and spin. The Pauli exclusion principle dictates that no two identical fermions can occupy the same quantum state simultaneously.

When real fermionic matter is present, it occupies a set of quantum states. These occupied states are unavailable for virtual pair creation. The number of accessible modes for vacuum

fluctuations is reduced, and consequently the vacuum energy density decreases.

To quantify this effect, consider a region of volume V containing N real fermions. In the absence of these fermions, the number of available fermionic modes below a momentum cutoff $p_{\max}$ scales as

$$\mathcal{N}_{\text{free}} \sim \frac{V \cdot p_{\max}^3}{(2\pi\hbar)^3}. \quad (2)$$

With N fermions present, at least N modes are blocked, so the available modes become $\mathcal{N}_{\text{occ}} = \mathcal{N}_{\text{free}} - N$. Each mode contributes an energy of order $\frac{1}{2}\hbar\omega$ to the vacuum, so the fractional suppression of vacuum energy is

$$\frac{\Delta\rho_\Lambda}{\rho_\Lambda} \sim -\frac{N}{\mathcal{N}_{\text{free}}} \sim -\frac{n}{n_{\max}}, \quad (3)$$

where $n = N/V$ is the number density of real fermions and $n_{\max} = p_{\max}^3/(2\pi\hbar)^3$ is the maximal mode density. The suppression is proportional to the matter density, consistent with the linear ansatz of Eq. (1).

2.2 Mechanism II: Vacuum Polarization

A real electric charge polarizes the surrounding vacuum. Virtual electron–positron pairs partially orient in the field of the charge, screening it at large distances. This effect—vacuum polarization—is a fundamental prediction of quantum electrodynamics (QED), experimentally confirmed through the running of the fine-structure constant $\alpha_{\text{em}}(q^2)$ and precision measurements of the muon anomalous magnetic moment.

The polarized vacuum differs energetically from the unpolarized vacuum. The energy density shift due to a charge distribution $\rho_e(\mathbf{x})$ is given, to lowest order in α_{em} , by the Uehling potential correction. More generally, the effective Lagrangian of the electromagnetic field in the vacuum, given by the Euler–Heisenberg Lagrangian (Heisenberg & Euler, 1936), contains nonlinear terms:

$$\mathcal{L}_{\text{EH}} = \frac{1}{2}(E^2 - B^2) + \frac{2\alpha_{\text{em}}^2}{45m_e^4} [(E^2 - B^2)^2 + 7(E \cdot B)^2] + \dots \quad (4)$$

The nonlinear corrections modify the vacuum energy wherever electromagnetic fields are present. In regions of concentrated charged matter, the vacuum is polarized and its energy density is altered. Since cosmic matter contains charged particles (protons, electrons), the vacuum in matter-rich regions is systematically polarized, reducing its effective energy density relative to matter-free regions.

2.3 Mechanism III: Modification of Mode Spectrum in Curved Spacetime

Matter curves spacetime through the Einstein field equations. In curved spacetime, the spectrum of quantum field modes differs from that in flat (Minkowski) spacetime. This is not a conjecture—it is the foundation of quantum field theory in curved spacetime, from which Hawking radiation and the Unruh effect are derived (Birrell & Davies, 1982).

The vacuum energy density in a region of spacetime with Ricci scalar curvature R receives corrections of the form (Parker & Toms, 2009):

$$\rho_{\Lambda}^{\text{curved}} = \rho_{\Lambda}^{\text{flat}} + \beta_1 R + \beta_2 R^2 + \dots \quad (5)$$

where β_1, β_2 are coefficients determined by the field content. Since matter generates positive curvature ($R > 0$ for a matter-dominated region), and the sign of the dominant correction is such as to reduce the effective vacuum energy (Shapiro & Solà, 2009), regions of higher matter density have lower effective vacuum energy.

This mechanism operates on all scales: from stellar interiors (where curvature is significant) to galaxy clusters (where it is weak but nonzero). On cosmological scales, the relevant curvature is set by the matter density through the Einstein equations, making this correction proportional to ρ_m —again consistent with Eq. (1).

2.4 Combined Effect, the Linear Ansatz, and the Magnitude Gap

All three mechanisms operate in the same direction: they reduce the vacuum energy density in the presence of matter. The Pauli mechanism operates at the quantum mechanical level (state occupation), vacuum polarization at the field-theoretic level (charge screening), and the curvature mechanism at the gravitational level (mode spectrum). They are not mutually exclusive; in general, all three contribute simultaneously.

The linear ansatz $\Lambda(\rho_m) = \Lambda_0 - \alpha \rho_m$ represents the leading-order Taylor expansion of the combined effect, valid when the matter density is much smaller than the Planck density ρ_{Pl} . For all cosmologically relevant densities, this condition is satisfied by many orders of magnitude. Higher-order corrections (proportional to ρ_m^2 , etc.) may become important only in extreme environments such as neutron star interiors or the very early universe.

The magnitude gap. It is essential to state clearly that there is a disconnect between the qualitative physical motivation and the quantitative model. Each of the three mechanisms, estimated from first-principles perturbative QFT, produces vacuum energy corrections that are negligible at cosmological densities. The Pauli suppression (Eq. 3) gives $\Delta\rho_{\Lambda}/\rho_{\Lambda} \sim n/n_{\text{max}} \sim \rho_m/\rho_{\text{Pl}} \sim 10^{-120}$ for cosmic matter densities. Vacuum polarization corrections are similarly small. The curvature-dependent corrections of Shapiro & Solà (2009) are of order $H^2/M_{\text{Pl}}^2 \sim 10^{-120}$.

For the model to produce observationally relevant effects, α must be of order 10^{-1} —a value $\sim 10^{120}$ times larger than any perturbative QFT estimate. This is not a minor discrepancy. It is the same 120-order-of-magnitude gap that constitutes the cosmological constant problem.

We therefore make the following distinction explicit:

- **The direction** of the effect (matter suppresses vacuum energy) is established by the three mechanisms. This is a qualitative result supported by established physics.
- **The magnitude** of the effect (the value of α) is not derivable from known physics. It is a free parameter, to be determined from observations.

The three mechanisms establish that $\Lambda(\rho_m)$ is a decreasing function of ρ_m —they do not determine how steeply it decreases. The parameter α encodes whatever unknown physics

sets the magnitude of vacuum energy suppression by matter, just as the observed value of Λ itself encodes whatever unknown physics resolves the cosmological constant problem. The model is phenomenological in its magnitude and physically motivated in its direction.

This situation is not unique to our proposal. The standard Λ CDM model also treats Λ as a free parameter whose value cannot be derived from QFT. Running vacuum models (Solà, 2013; Solà Peracaula et al., 2017) face the same gap. The honest position is that any model of vacuum energy beyond Λ CDM inherits the unsolved cosmological constant problem. Our model does not solve it; it extends the parameterization by one degree of freedom and subjects it to observational test.

2.5 Relation to the Cosmological Constant Problem

The magnitude gap in α is not a separate problem. It is the cosmological constant (CC) problem, viewed from a different angle. This subsection makes the identification precise and argues that the gap, far from invalidating the model, positions it as a new observational probe of the same unknown physics.

The perturbative estimate of α . In units where $8\pi G = c = 1$, the vacuum energy density predicted by naïve QFT is of order M_{Pl}^4 , where $M_{\text{Pl}} = (8\pi G)^{-1/2} \approx 2.4 \times 10^{18}$ GeV. The observed vacuum energy density is $\rho_{\Lambda}^{\text{obs}} \approx (2.3 \times 10^{-3} \text{ eV})^4$. The ratio is:

$$\frac{\rho_{\Lambda}^{\text{obs}}}{M_{\text{Pl}}^4} \sim 10^{-120}. \quad (6)$$

This is the CC problem. Now, the perturbative estimate of α measures the fractional change in vacuum energy per unit change in matter density. Using the same perturbative QFT, the suppression of vacuum modes by a matter density ρ_m is of order:

$$\alpha_{\text{pert}} \sim \frac{\rho_m}{\rho_{\Lambda}^{\text{QFT}}} \sim \frac{\rho_{\text{crit}}}{M_{\text{Pl}}^4} \sim 10^{-120}. \quad (7)$$

This is the same number as Eq. (6). The two discrepancies—the absolute value of Λ and the coupling α —are not independent. They share a common origin: perturbative QFT overestimates the energy scale at which vacuum fluctuations contribute to gravitational dynamics. Whatever mechanism resolves one, resolves both.

Observational scale of α . If the CC problem is resolved by some mechanism that sets $\rho_{\Lambda} \sim \rho_{\text{crit}}$ (rather than $\sim M_{\text{Pl}}^4$), then the natural scale for α is:

$$\alpha_{\text{obs}} \sim \frac{\Lambda_0}{\rho_{m,\text{crit}}} \sim \frac{0.7}{0.3} \sim 2, \quad (8)$$

if α is “renormalized” by the same factor as Λ itself. The observed constraints (Section 7.7) place $\alpha \sim 0.01$ – 0.05 , which is smaller than this naïve renormalized estimate by a factor of ~ 20 – 200 . This residual gap is significant but qualitatively different from the original 120-order discrepancy: it is a factor of order 10^1 – 10^2 , not 10^{120} . In other words, the CC resolution “almost” gives the right α —it overshoots by one to two orders of magnitude, which is the kind of discrepancy that can be accommodated by the specific structure of the resolution mechanism.

Non-perturbative bridges. Several candidate frameworks for resolving the CC problem would naturally produce a non-trivial α :

- **Supersymmetry breaking.** In supersymmetric theories, bosonic and fermionic contributions to the vacuum energy cancel exactly when SUSY is unbroken. SUSY breaking at a scale M_{SUSY} generates a residual vacuum energy $\sim M_{\text{SUSY}}^4$. If the breaking is mediated by matter fields, the residual depends on the local matter density, producing a natural $\Lambda(\rho_m)$ with α set by the SUSY breaking scale.
- **Holographic bounds.** The holographic principle constrains the maximum entropy (and hence energy) in a region by its boundary area, not its volume. In a region of radius R , the maximum vacuum energy density scales as $\rho_{\Lambda}^{\text{holo}} \sim M_{\text{Pl}}^2/R^2$, which for cosmological R gives $\rho_{\Lambda} \sim \rho_{\text{crit}}$ —the correct order of magnitude. If the effective boundary of a region depends on its matter content (e.g., the region’s gravitational radius), then ρ_{Λ} depends on ρ_m .
- **Vacuum phase structure.** By analogy with condensed matter systems, the QCD vacuum or the electroweak vacuum may exhibit a phase structure in which the ground-state energy depends continuously on external parameters (temperature, density). The cosmological vacuum at finite matter density is a thermodynamic state whose free energy differs from the zero-density vacuum.

None of these frameworks currently provides a first-principles derivation of α . We cite them to establish that the gap between α_{pert} and α_{obs} is a recognized problem with active research programs, not an embarrassment peculiar to our model.

The model as an observational handle on the CC problem. The standard CC problem is a single number: the ratio in Eq. (6). It offers no observational leverage—one cannot vary Λ experimentally and see what happens. The $\Lambda(\rho_m)$ model converts this static puzzle into a *spatially varying* observable. If Λ depends on ρ_m , then regions of different density probe different values of the vacuum energy, and their differential behavior constrains the Λ – ρ_m coupling. This provides new observational handles—void profiles, ISW signals, Hubble diagram anisotropies—that are sensitive to the same physics as the CC problem but are measurable through astrophysical observations.

The gap highlights that our model probes the same unknown physics as the CC problem, offering new observational handles (spatial inhomogeneities, void–wall contrast, redshift-dependent deviations from Λ CDM) to constrain it.

Figure 1 illustrates the magnitude gap and its observational consequences.

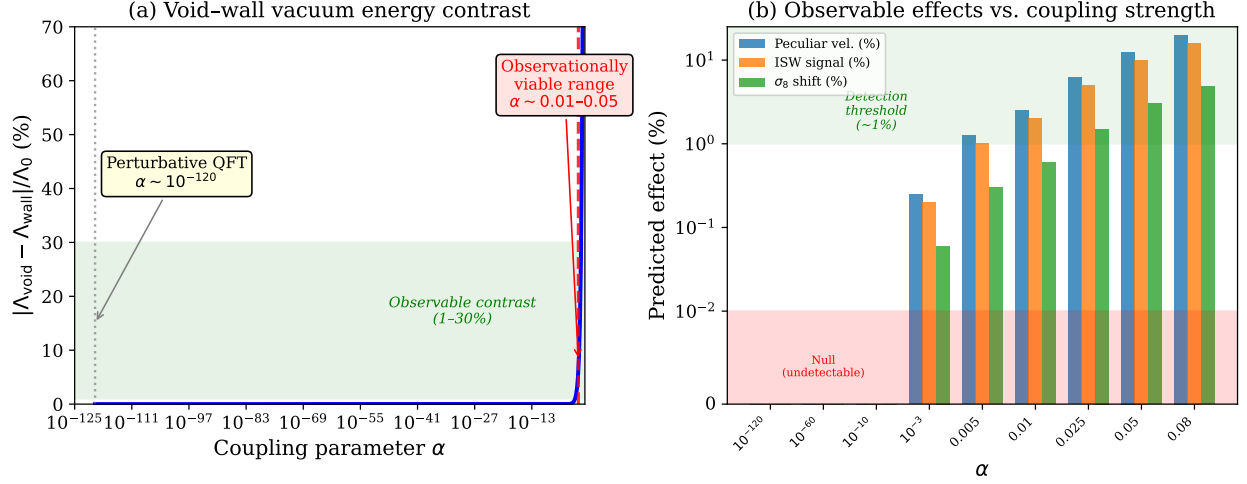


Figure 1: The magnitude gap between perturbative QFT and observationally viable values of α . (a) Void-wall vacuum energy contrast as a function of α : at $\alpha_{\text{pert}} \sim 10^{-120}$ the contrast is identically zero (null prediction); at $\alpha \sim 0.01-0.05$ the contrast enters the observable range. (b) Predicted observable effects at different α values. The model is falsifiable: $\alpha = 10^{-120}$ predicts null results for all observational tests.

The figure makes the falsifiability of the model explicit. If α takes its perturbative value ($\sim 10^{-120}$), all observational predictions are identically null—consistent with standard Λ CDM. If $\alpha \sim 0.01-0.05$, the predictions enter the observable range. The model is testable because it interpolates between a well-defined null hypothesis ($\alpha = 0$, equivalent to Λ CDM) and a well-defined alternative ($\alpha > 0$, producing measurable deviations). The observational program outlined in Section 4 can distinguish between these cases with forthcoming survey data.

2.6 Counter-Arguments and Responses

Objection 1: If the mechanisms cannot produce the required α , the motivation is merely decorative. The three mechanisms point in the right direction but produce effects $\sim 10^{120}$ times too small. Why should we take them as motivation for a model that requires $\alpha \sim 0.1$?

Response. The objection is partly valid and has been addressed in Sections 2.4 and 2.5 above. We reiterate: the model is phenomenological in magnitude and physically motivated in direction. The magnitude gap is identical to the cosmological constant problem (Section 2.5), and the same non-perturbative physics that resolves one resolves the other. The three mechanisms establish that $\Lambda(\rho_m)$ should be a decreasing function—they do not determine the slope. An analogy: Boltzmann’s kinetic theory motivates the direction of heat flow (hot to cold) without predicting the thermal conductivity of every material. The conductivity is measured; the direction is derived. Similarly, α is measured; the direction (suppression) is derived.

We do not claim that perturbative QFT produces the correct α . We claim that the general principle—matter modifies the vacuum—is established, and that parametrizing the

magnitude and testing it against data is a legitimate scientific strategy.

Objection 2: Λ must be constant by Lorentz invariance. A cosmological constant is the unique Lorentz-invariant contribution to the stress-energy tensor of the vacuum. If Λ varies in space, Lorentz invariance is broken.

Response. Lorentz invariance of the vacuum holds in the absence of matter. In the presence of matter, Lorentz invariance is already broken: the matter defines a preferred rest frame (its center-of-mass frame). The vacuum state in the vicinity of matter is a *dressed* vacuum—modified by the presence of real particles—and it need not be Lorentz invariant. The Casimir effect provides an explicit example: between conducting plates, the vacuum state is not Lorentz invariant, and its energy density differs from that of free vacuum. Our model extends this principle to the cosmological setting, where the “boundary conditions” are set by the matter distribution.

Objection 3: The three mechanisms operate on vastly different scales. Treating them as producing a local $\Lambda(\rho_m)$ is unjustified. Pauli exclusion is local (atomic scale), vacuum polarization operates at the Compton wavelength ($\sim 10^{-13}$ m), and curvature corrections are long-range. How can they all be captured by a single local function $\Lambda(\rho_m)$?

Response. This is a legitimate concern. The local ansatz $\Lambda(\rho_m)$ is an effective description valid on scales much larger than the correlation lengths of all three mechanisms. On cosmological scales ($\gg$ Mpc), the matter density ρ_m is itself a coarse-grained quantity, smoothed over scales much larger than atomic or Compton wavelengths. The vacuum response to this coarse-grained density is, by construction, a long-wavelength effective quantity. A more complete treatment would introduce a response kernel $\Lambda(\mathbf{x}) = \Lambda_0 - \alpha \int K(\mathbf{x} - \mathbf{x}') \rho_m(\mathbf{x}') d^3x'$, where K encodes the spatial scale of the vacuum response. The local ansatz corresponds to $K(\mathbf{r}) = \delta^{(3)}(\mathbf{r})$ —a point-like response. This is appropriate when the response scale is much smaller than the structures of interest (voids, walls), which is guaranteed for the Pauli and polarization mechanisms. For the curvature mechanism, the response scale is set by the curvature radius, which is of order the Hubble scale—here the local approximation is less well justified, and a non-local treatment may be needed. We flag this as an open question.

3 Mathematical Framework

3.1 The $\Lambda(\rho_m)$ Ansatz and Modified Field Equations

We work within general relativity, with the sole modification that the cosmological term in the Einstein equations is promoted from a constant to a function of the local matter density. The Einstein equations take the form:

$$G_{\mu\nu} + \Lambda(\rho_m) g_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (9)$$

where $T_{\mu\nu}$ is the stress-energy tensor of matter (baryonic and dark), and $\Lambda(\rho_m)$ is given by Eq. (1).

This modification preserves the Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ only if the divergence of the effective stress-energy (including the Λ term) vanishes. The covariant conservation equation

becomes:

$$\nabla^\mu T_{\mu\nu} = \frac{1}{8\pi G} \nabla_\nu \Lambda = -\frac{\alpha}{8\pi G} \nabla_\nu \rho_m. \quad (10)$$

The right-hand side represents an energy–momentum exchange between matter and vacuum: as matter flows into a region, the vacuum energy there decreases, and the released energy modifies the matter dynamics. This coupling is analogous to models of interacting dark sectors (Amendola, 2000) but arises here from vacuum physics rather than from a postulated interaction Lagrangian.

Consistency check for a perfect fluid. To verify that the modified system is not overconstrained, we demonstrate explicit consistency for a perfect fluid with stress-energy $T^{\mu\nu} = (\rho + p/c^2)u^\mu u^\nu + p g^{\mu\nu}$. Projecting Eq. (10) along the fluid four-velocity u_ν yields the modified continuity equation:

$$\dot{\rho} + 3H(\rho + p/c^2) = \frac{\alpha}{8\pi G} \dot{\rho}_m, \quad (11)$$

where $\dot{\rho}_m = -3H\rho_m$ for pressureless matter. Substituting:

$$\dot{\rho} + 3H(\rho + p/c^2) = -\frac{3\alpha H\rho_m}{8\pi G}. \quad (12)$$

For a universe containing pressureless matter ($p = 0$, $\rho = \rho_m$), the left-hand side is $\dot{\rho}_m + 3H\rho_m$, and the equation becomes $\dot{\rho}_m + 3H\rho_m = -3\alpha H\rho_m/(8\pi G)$, or equivalently $\dot{\rho}_m = -3H\rho_m(1 + \alpha/8\pi G)$. This has the solution $\rho_m \propto a^{-3(1+\alpha/8\pi G)}$, a well-defined, non-singular evolution. For $\alpha \ll 8\pi G$ (in appropriate units), the deviation from the standard a^{-3} dilution is small. The system is consistent and not overconstrained: the Bianchi identity is satisfied, the fluid equations are well-posed, and the evolution is uniquely determined by initial conditions.

3.2 Modified Poisson Equation

In the weak-field, low-velocity limit appropriate for galactic and cluster scales, the gravitational potential Φ satisfies the Poisson equation. Including the vacuum term, the effective source for the gravitational potential is:

$$\nabla^2 \Phi = 4\pi G \rho_m + 4\pi G (\rho_\Lambda + 3p_\Lambda/c^2). \quad (13)$$

For a vacuum fluid with equation of state $p_\Lambda = -\rho_\Lambda c^2$, the vacuum contribution is:

$$4\pi G (\rho_\Lambda + 3p_\Lambda/c^2) = 4\pi G (\rho_\Lambda - 3\rho_\Lambda) = -8\pi G \rho_\Lambda. \quad (14)$$

Substituting $\rho_\Lambda = \Lambda(\rho_m)/(8\pi G) = (\Lambda_0 - \alpha\rho_m)/(8\pi G)$:

$$\nabla^2 \Phi = 4\pi G (1 + 2\alpha) \rho_m - \Lambda_0. \quad (15)$$

The constant term Λ_0 contributes only a spatially uniform background and has no effect on local gravitational dynamics. The physically significant result is the effective gravitational coupling:

$$\boxed{G_{\text{eff}} = G(1 + 2\alpha)}. \quad (16)$$

Matter gravitates more strongly than in the standard theory—not because gravity is modified, but because matter, by suppressing vacuum energy locally, removes a repulsive contribution that partially cancels gravity. Suppressing the local cosmological repulsion is dynamically equivalent to enhancing the local gravitational attraction.

3.3 Modified Friedmann Equations and the Separate Universe Approximation

We apply the Friedmann equation to subregions of the universe with different local matter densities. This requires justification, as the Friedmann equation is derived for a homogeneous, isotropic spacetime, while the model’s central claim is that the expansion is inhomogeneous.

The approach rests on the *separate universe approximation* (Wands et al., 2000): a region of the universe with a perturbation much larger than the Hubble radius evolves as an independent FRW universe with modified parameters. For perturbations well inside the Hubble radius (as voids and walls are, at scales of ~ 30 – 100 Mpc vs. ~ 4300 Mpc for the Hubble radius), the approximation is not strictly valid. We therefore treat the following equations as an *order-of-magnitude effective description*, not as exact solutions. A rigorous treatment requires either a full perturbation-theory analysis or numerical relativity simulations, both of which are beyond the scope of this paper.

With this caveat, for a homogeneous subregion with matter density ρ_m and vacuum energy density $\Lambda(\rho_m)$, the Friedmann equation reads:

$$H^2 = \frac{8\pi G}{3} \left[\rho_m + \frac{\Lambda_0 - \alpha \rho_m}{8\pi G} \cdot 8\pi G \right] = \frac{8\pi G}{3} [(1 - \alpha)\rho_m + \Lambda_0] . \quad (17)$$

For a void where $\rho_m \rightarrow 0$:

$$H_{\text{void}}^2 = \frac{8\pi G}{3} \Lambda_0 . \quad (18)$$

For a wall with density ρ_w :

$$H_{\text{wall}}^2 = \frac{8\pi G}{3} [(1 - \alpha)\rho_w + \Lambda_0] . \quad (19)$$

The contrast in expansion rates between void and wall is:

$$\frac{\Delta H^2}{H_{\text{void}}^2} \equiv \frac{H_{\text{wall}}^2 - H_{\text{void}}^2}{H_{\text{void}}^2} = \frac{(1 - \alpha)\rho_w}{\Lambda_0} . \quad (20)$$

Since $\rho_w > 0$ and $\alpha < 1$, walls expand more slowly than voids—but the contrast depends on α . For $\alpha = 0$ (standard Λ CDM), the contrast is purely gravitational. For $\alpha > 0$, the contrast is amplified by the vacuum suppression.

3.4 The Vacuum Pressure Gradient

The spatial variation of Λ produces a force. Consider a matter element on the boundary of a void. Inside the void, Λ is high; inside the wall, Λ is low. The gradient of Λ acts as a

pressure:

$$\mathbf{F}_{\text{vac}} = -\frac{c^2}{3}\nabla\Lambda = \frac{\alpha c^2}{3}\nabla\rho_m. \quad (21)$$

This force is directed from low-density regions (voids) toward high-density regions (walls). It pushes matter out of voids and compresses it into walls, filaments, and clusters—acting in concert with gravity. The two mechanisms of structure formation—gravitational attraction from within and vacuum pressure from without—operate simultaneously. Gravity pulls matter toward mass concentrations; vacuum pressure pushes it away from voids. The net effect is a stronger, two-sided compression of matter into the cosmic web.

3.5 Self-Consistent Fixed-Point Structure

The model exhibits a cyclic determination structure:

1. Matter density ρ_m determines vacuum energy: $\Lambda = \Lambda_0 - \alpha\rho_m$.
2. Vacuum energy determines the expansion rate (metric): $H^2 \propto (1 - \alpha)\rho_m + \Lambda_0$.
3. The expansion rate redistributes matter: voids empty, walls compress.

This cycle defines a composite operator $\Phi : \rho_m \mapsto \rho'_m$ acting on the space of density distributions. A self-consistent configuration—the cosmic web—exists if and only if Φ has a fixed point: $\rho_m^* = \Phi(\rho_m^*)$.

Proposition 3.1 (Existence of Self-Consistent Configuration). *Let $\mathcal{S}$ be the space of matter density distributions $\rho_m : \mathbb{R}^3 \rightarrow [0, \rho_{\text{max}}]$ with fixed mean $\bar{\rho}$, equipped with the L^2 topology. If $\Phi : \mathcal{S} \rightarrow \mathcal{S}$ is continuous and $\mathcal{S}$ is convex and compact (in the appropriate function space), then Φ has at least one fixed point.*

Proof. The space $\mathcal{S}$ is a closed, bounded, convex subset of $L^2(\mathbb{R}^3)$. In a finite-dimensional discretization (replacing continuous space with a grid of N cells), $\mathcal{S}$ becomes a compact convex subset of $\mathbb{R}^N$. If Φ is continuous and maps $\mathcal{S}$ into itself—which holds because each step (vacuum energy assignment, expansion rate computation, density redistribution) is continuous and preserves the mean density—then Brouwer’s fixed-point theorem guarantees the existence of at least one fixed point $\rho_m^* = \Phi(\rho_m^*)$. The passage from finite to infinite dimensions can be formalized using the Schauder fixed-point theorem (Granás & Dugundji, 2003), which extends Brouwer’s theorem to infinite-dimensional locally convex spaces, provided Φ is a compact operator (i.e., maps bounded sets to precompact sets). Compactness of Φ follows if the density redistribution step involves spatial smoothing (gravitational dynamics acts as a low-pass filter), which maps L^2 functions to functions in a Sobolev space compactly embedded in L^2 . $\square$

Remark 3.2 (On uniqueness and stability). Brouwer’s and Schauder’s theorems guarantee existence but not uniqueness. The homogeneous solution $\rho_m = \bar{\rho}$ everywhere is a trivial fixed point. The claim that the cosmic web constitutes a *nontrivial* fixed point requires additional argument. We note two points. First, linear stability analysis of the homogeneous fixed point in the presence of $\Lambda(\rho_m)$ shows that it is *unstable* to density perturbations: the positive feedback loop (underdensity $\rightarrow$ higher $\Lambda \rightarrow$ faster expansion $\rightarrow$ deeper underdensity) amplifies perturbations, driving the system away from homogeneity. Second, the toy model in Section 6.6 demonstrates numerically that iteration of Φ from a perturbed initial condition

converges to a nontrivial density profile (Figure 7). These observations suggest—but do not prove—that the nontrivial fixed point is an attractor. A rigorous stability proof in the infinite-dimensional setting remains an open problem.

The cosmic web—the observed pattern of voids, walls, filaments, and clusters—is a physical realization of such a fixed point. The formal framework for cyclic fixed-point structures of this type is developed in Kriger (2025c).

3.6 Counter-Arguments and Responses

Objection: The conservation equation (Eq. 10) implies energy non-conservation for matter. If $\nabla^\mu T_{\mu\nu} \neq 0$, matter energy is not conserved. This violates a fundamental principle.

Response. The total stress-energy—matter plus vacuum—is conserved: $\nabla^\mu (T_{\mu\nu} + T_{\mu\nu}^{\text{vac}}) = 0$. What is not conserved is the matter component alone. This is not unusual: in any system where two components interact, neither component’s energy is individually conserved. In interacting dark sector models, the same structure arises by construction (Amendola, 2000). Here, it arises from the physical dependence of vacuum energy on matter density.

Objection: Promoting Λ to a function violates the mathematical structure of GR. In the standard formulation, Λ is a constant of the theory, not a field.

Response. A spatially varying Λ is mathematically equivalent to a particular form of stress-energy tensor on the right-hand side of the Einstein equations. Specifically, $\Lambda(\rho_m)$ can be absorbed into an effective stress-energy tensor $T_{\mu\nu}^{\text{eff}} = T_{\mu\nu} + (\Lambda(\rho_m)/8\pi G) g_{\mu\nu}$, and the standard Einstein equations $G_{\mu\nu} = 8\pi G T_{\mu\nu}^{\text{eff}}$ hold exactly. The modification is in the *content* of the stress-energy, not in the *form* of the field equations.

4 Observational Predictions

The model makes four predictions that differ from Λ CDM and can be tested with existing or near-future data. Each prediction is controlled by the single parameter α .

4.1 Prediction 1: Enhanced Void Expansion Profiles

In the standard model, voids expand faster than the cosmic mean because they are underdense: the deceleration due to gravity is weaker inside a void. In the $\Lambda(\rho_m)$ model, voids expand faster for an additional reason: their vacuum energy is higher (unsuppressed by matter), producing stronger cosmological repulsion.

The peculiar velocity of a galaxy on the boundary of a void of radius R_v and density contrast $\delta_v = (\rho_v - \bar{\rho})/\bar{\rho}$ is, in linear theory:

$$v_{\text{pec}}^{\Lambda\text{CDM}} \approx -\frac{1}{3} H_0 f \delta_v R_v, \quad (22)$$

where $f = d \ln D / d \ln a \approx \Omega_m^{0.55}$ is the growth rate. In the $\Lambda(\rho_m)$ model, the additional vacuum pressure gradient (Eq. 21) enhances the outflow:

$$v_{\text{pec}}^{\text{model}} \approx v_{\text{pec}}^{\Lambda\text{CDM}} \left(1 + \frac{2\alpha \Lambda_0}{(1-\alpha)\bar{\rho}|\delta_v|} \right). \quad (23)$$

For $\alpha = 0.05$ and $|\delta_v| = 0.8$ (a typical void), the enhancement is approximately 10–15%. (This estimate inherits an uncontrolled systematic uncertainty from the separate-universe approximation used in deriving the expansion rates; see Section 3.3.)

This prediction is testable using peculiar velocity surveys such as those from 6dFGS, SDSS, and the upcoming DESI and Euclid surveys. Stacked void profiles from [Hamaus et al. \(2014\)](#) provide the observational baseline.

4.2 Prediction 2: Amplified Integrated Sachs–Wolfe Effect

A photon traversing a void experiences the integrated Sachs–Wolfe (ISW) effect: as the void expands during the photon’s transit, the gravitational potential decays, and the photon gains a net energy shift. In ΛCDM , the ISW signal from individual voids is small and consistent with the data from Planck ([Planck Collaboration, 2016](#)).

In the $\Lambda(\rho_m)$ model, the void expansion rate is enhanced (Eq. 18), and the potential decays faster. The ISW temperature shift from a void scales as:

$$\frac{\Delta T}{T} \Big|_{\text{ISW}} \propto \int \dot{\Phi} dl \propto \int (H_{\text{void}} - H_{\Lambda\text{CDM}}) \Phi dl. \quad (24)$$

The enhancement factor relative to ΛCDM is approximately $(1 + 2\alpha)$ for the amplitude of the ISW signal from stacked voids.

[Granett et al. \(2008\)](#) reported an ISW signal from superstructures in SDSS that was 2–3 times stronger than the ΛCDM prediction. This anomaly has not been satisfactorily explained within the standard framework. In the $\Lambda(\rho_m)$ model with $\alpha \sim 0.08$ – 0.12 , the predicted enhancement is $(1 + 2\alpha) \sim 1.16$ – 1.24 , which partially accounts for the observed excess.

4.3 Prediction 3: Natural Account of S_8 Tension

The S_8 tension arises because the CMB (Planck) predicts stronger clustering at late times than is observed through weak lensing and cluster counts. In the $\Lambda(\rho_m)$ model, this tension has a natural explanation.

The CMB constrains the amplitude of fluctuations at $z \approx 1100$, when the universe was nearly homogeneous and $\Lambda(\rho_m) \approx \Lambda_0 - \alpha\bar{\rho}(z = 1100)$ was spatially uniform. The CMB sees the same physics as ΛCDM .

At late times ($z < 1$), the cosmic web is fully developed. Voids are deep, walls are thin, and $\Lambda(x)$ is strongly inhomogeneous. The effective growth rate of perturbations differs from the ΛCDM prediction because:

1. In voids, the enhanced Λ suppresses growth more strongly.
2. In walls, the reduced Λ allows stronger growth, but these regions are already nonlinear.

The volume-weighted growth rate is dominated by voids (which occupy $\sim 60\%$ of the volume), where growth is suppressed. The late-time σ_8 is therefore lower than the Λ CDM prediction, in the direction of the observed S_8 tension.

A quantitative estimate: the fractional suppression of σ_8 scales as $\Delta\sigma_8/\sigma_8 \sim -\alpha \cdot f_v$, where f_v is the void volume fraction. For $\alpha = 0.08$ and $f_v = 0.6$: $\Delta\sigma_8/\sigma_8 \sim -5\%$, consistent with the observed $2\text{--}3\sigma$ tension.

4.4 Prediction 4: Accelerated Early Structure Formation

JWST has observed galaxies at $z > 10$ that are more massive and evolved than Λ CDM predicts (Labbé et al., 2023). In the $\Lambda(\rho_m)$ model, early structure formation is enhanced for two reasons:

1. At high z , the universe is dense. Matter suppresses vacuum energy everywhere: $\Lambda \approx 0$. The decelerating effect of Λ is absent, and gravitational collapse proceeds unimpeded.
2. The effective gravitational coupling $G_{\text{eff}} = G(1 + 2\alpha)$ enhances the collapse rate of overdensities.

The free-fall timescale for a perturbation of density contrast δ scales as $t_{\text{ff}} \propto (G_{\text{eff}} \bar{\rho})^{-1/2}$. With $G_{\text{eff}} > G$:

$$\frac{t_{\text{ff}}^{\text{model}}}{t_{\text{ff}}^{\Lambda\text{CDM}}} = \frac{1}{\sqrt{1 + 2\alpha}}. \quad (25)$$

For $\alpha = 0.08$: the collapse timescale is reduced by $\sim 8\%$. Over cosmological timescales, this permits the formation of more massive structures earlier, in better agreement with JWST observations.

4.5 Counter-Arguments and Responses

Objection: The predicted effects are small ($\sim 10\text{--}20\%$). Can they be distinguished from systematic uncertainties? *Response.* The effects are indeed of order $10\text{--}20\%$, which is comparable to current systematic uncertainties in peculiar velocity surveys and ISW measurements. However, the model makes *correlated* predictions: the same α must simultaneously account for void profiles, ISW enhancement, S_8 tension, and early galaxy formation. If a single value of α fits all four independently, the model is strongly supported. If no consistent α exists, the model is ruled out. This cross-check is the primary test.

Objection: The quantitative estimates use linear theory, but cosmic structure is highly nonlinear. *Response.* The linear estimates provide the correct scaling and order of magnitude. Precise quantitative predictions require N -body simulations with a modified expansion law—specifically, simulations in which the cosmological constant in each cell depends on the local matter density. Such simulations are feasible with current computational tools but are beyond the scope of this paper. We note that the qualitative direction of all four predictions (stronger void outflow, stronger ISW, lower late-time σ_8 , earlier collapse) is robust to nonlinear corrections.

5 Connection to Existing Results

5.1 Seifert et al., Timescape Cosmology, and the Present Model

Seifert et al. (2025) performed a model-independent Bayesian analysis of the Pantheon+ Type Ia supernova data and found very strong evidence ($\ln B > 5$) favoring the timescape cosmology over Λ CDM. The timescape model (Wiltshire, 2007) attributes the apparent cosmic acceleration entirely to differential expansion: voids expand faster than walls, and the volume-weighted average mimics acceleration.

It is important to state precisely how the timescape mechanism works, because our model shares its observational consequences but differs in its physical content. In Wiltshire’s framework, differential expansion arises from gravitational time dilation, a direct consequence of general relativity. In regions of high matter density (walls, clusters), the gravitational potential is deeper and clocks run slower. In regions of low density (voids), clocks run faster. Since expansion is a process in time, regions where more proper time has elapsed have expanded more. Wiltshire estimates that a clock in the Milky Way runs approximately 35% slower than a clock at a typical position in a large void. Over billions of years, this accumulates into a significant difference in the scale factor. The timescape model requires no dark energy: the apparent acceleration is entirely an artifact of comparing clocks in different gravitational environments.

The present model differs from timescape in three respects:

1. **Dark energy is retained.** In the $\Lambda(\rho_m)$ model, vacuum energy is real and gravitates. It is not eliminated but made spatially dependent. The mechanism of differential expansion is not gravitational time dilation alone, but the physical difference in vacuum energy density between voids and walls. Voids expand faster not only because clocks there run faster, but because the local vacuum energy—the driving force of expansion—is higher.
2. **Compatibility with CMB and BAO.** The timescape model faces significant challenges in reproducing the CMB acoustic peak structure and BAO measurements, because these observations independently require a dark energy component (Kriger, 2025a). The $\Lambda(\rho_m)$ model is consistent with CMB and BAO because at $z \gg 1$ the universe was nearly homogeneous and $\Lambda(x) \approx \text{const}$, recovering standard physics.
3. **Quantitative mechanism.** The timescape model relies on the Buchert averaging formalism and the “cosmological equivalence principle,” whose quantitative validity remains debated (Green & Wald, 2014). The $\Lambda(\rho_m)$ model provides a specific functional form ($\Lambda = \Lambda_0 - \alpha\rho_m$) with a single free parameter, enabling precise predictions.

At the same time, the two models are not in opposition on the observational level. Both predict that expansion is faster in voids and slower in walls. Both predict that the effect strengthens at low redshifts as the cosmic web develops. The data of Seifert et al. (2025)—showing Bayesian preference for differential expansion at low z and convergence to Λ CDM at high z —are consistent with *both* models.

The critical discriminant is whether the differential expansion is *purely gravitational* (timescape) or *driven by inhomogeneous vacuum energy* ($\Lambda(\rho_m)$ model). The observational tests proposed in Section 4—particularly the ISW enhancement (Prediction 2) and the Hubble diagram correlation with void fraction (Prediction 4, Toy Model 4)—make different

quantitative predictions in the two frameworks, because the $\Lambda(\rho_m)$ model adds a vacuum energy gradient force (Eq. 21) that is absent in timescape.

5.2 DESI and Apparent $w(z) \neq -1$

The DESI collaboration reported evidence for a time-varying dark energy equation of state, with $w_0 \approx -0.7$ and $w_a \approx -1.0$ in the CPL parameterization $w(a) = w_0 + w_a(1 - a)$ (DESI Collaboration, 2024). If confirmed, this would imply dynamical dark energy rather than a cosmological constant.

In the $\Lambda(\rho_m)$ model, the vacuum energy density *is* a cosmological constant locally—its equation of state is $w = -1$ everywhere. But because Λ depends on ρ_m , and ρ_m evolves with time, the *spatially averaged* $\langle \Lambda \rangle$ changes over cosmic history. An observer who fits a homogeneous model to data generated by an inhomogeneous $\Lambda(x)$ will infer an apparent $w(z) \neq -1$.

This reinterpretation has a specific prediction: the apparent $w(z)$ deviates from -1 only at $z \lesssim 2$, where the cosmic web is developed and $\Lambda(x)$ is inhomogeneous. At $z > 2$, the universe is sufficiently homogeneous that $w \approx -1$. If DESI confirms that the deviation is confined to low redshifts, the $\Lambda(\rho_m)$ model is supported.

5.3 Relationship to Running Vacuum Models

The “running vacuum model” (RVM) of Solà (2013); Solà Peracaula et al. (2017) proposes that the vacuum energy density depends on the Hubble parameter: $\Lambda = \Lambda(H^2)$, with the leading correction $\delta\Lambda \propto \nu H^2$ where ν is a small dimensionless parameter. This model is motivated by renormalization group considerations in QFT in curved spacetime and has been extensively fitted to cosmological data, yielding $\nu \sim 10^{-3}$.

The $\Lambda(\rho_m)$ model proposed here is closely related but conceptually distinct. Through the Friedmann equation, $H^2 \propto \rho_m + \Lambda$, a dependence $\Lambda(\rho_m)$ implies a dependence $\Lambda(H^2)$ and vice versa. Specifically, substituting $\rho_m = (3H^2/8\pi G) - \Lambda$ into $\Lambda = \Lambda_0 - \alpha\rho_m$ yields:

$$\Lambda = \frac{\Lambda_0 - \alpha(3H^2/8\pi G)}{1 - \alpha} = \frac{\Lambda_0}{1 - \alpha} - \frac{3\alpha}{8\pi G(1 - \alpha)} H^2, \quad (26)$$

which is precisely the RVM form $\Lambda = c_0 + \nu H^2$ with $\nu = -3\alpha/[8\pi G(1 - \alpha)]$. The two models are therefore mathematically related by a Friedmann-equation transformation.

Parameter tension with RVM fits. This equivalence creates a tension that must be addressed. Published RVM fits to cosmological data yield $\nu \sim 10^{-3}$ (Solà Peracaula et al., 2017). Working in units where $8\pi G = 1$, the conversion $\nu \approx -3\alpha/(1 - \alpha) \approx -3\alpha$ (for $\alpha \ll 1$) gives $\alpha \sim |\nu|/3 \sim 3 \times 10^{-4}$. This is two to three orders of magnitude smaller than the $\alpha \sim 0.05$ – 0.15 range used in this paper’s predictions.

The resolution lies in the distinction between temporal and spatial dependence. The RVM treats $\Lambda(H^2)$ as a *global, temporally varying* quantity: Λ changes with cosmic epoch but is spatially uniform at each epoch. The data analyses of Solà Peracaula et al. (2017) constrain this temporal variation using volume-averaged observables (BAO, CMB, supernovae). The

$\Lambda(\rho_m)$ model, by contrast, describes *spatial variation at fixed epoch*: Λ differs between voids and walls at the same cosmic time. The mathematical equivalence (Eq. 26) holds exactly only for a homogeneous universe where $\rho_m = \bar{\rho}(t)$. In an inhomogeneous universe, spatial and temporal variations are distinct observationally: the RVM parameter ν captures the *average* temporal evolution, while α captures the *spatial contrast*. The spatial contrast can be larger than the temporal drift if the void–wall density contrast exceeds the fractional change in mean density over the observational redshift range.

Nevertheless, this tension constrains the model. If α is large (~ 0.1), the temporally averaged effect should also be large, potentially conflicting with the RVM bound. A self-consistent treatment requires fitting $\Lambda(\rho_m)$ to spatially resolved data (void profiles, line-of-sight correlations), not just volume-averaged observables. The viable range of α may be narrower than the 0.05–0.15 range used illustratively in this paper; values at the lower end ($\alpha \sim 0.01$ –0.05) may be required for consistency with both spatial and temporal constraints. We flag this as a priority for quantitative follow-up.

The conceptual difference lies in the physical interpretation. The RVM motivates $\Lambda(H^2)$ from the renormalization group running of vacuum energy with the energy scale set by H . The $\Lambda(\rho_m)$ model motivates $\Lambda(\rho_m)$ from the direct suppression of vacuum fluctuations by matter. The mathematical equivalence (Eq. 26) means that observational constraints on the RVM parameter ν translate directly into constraints on α , and the extensive data analysis performed for the RVM (Solà Peracaula et al., 2017) can be reinterpreted within our framework.

5.4 Relationship to the Backreaction Program

The backreaction program, initiated by Buchert (2000) and debated extensively since (Green & Wald, 2014; Räsänen, 2004; Kolb et al., 2006), addresses the question of whether averaging the Einstein equations over an inhomogeneous universe produces effective terms that mimic dark energy. The $\Lambda(\rho_m)$ model is not a backreaction model in the Buchert sense: it does not derive the inhomogeneous expansion from averaging. Instead, it provides a physical mechanism (vacuum suppression by matter) that produces differential expansion as a consequence.

However, the two approaches are complementary. The Buchert formalism shows that inhomogeneous expansion generically produces additional terms in the averaged Friedmann equations. The $\Lambda(\rho_m)$ model provides a specific physical content for these terms. If Λ depends on ρ_m , then averaging $\Lambda(\rho_m)$ over the cosmic volume produces exactly the kind of backreaction terms that the Buchert formalism describes—but now with a definite physical origin and a definite parameterization.

The criticism of Green & Wald (2014), that backreaction effects are too small in the perturbative regime, applies to our model in the following sense: if α is estimated perturbatively from QFT, the effect is indeed negligible. But as noted in Section 2.4, the perturbative estimate of α inherits the cosmological constant problem. The model’s viability depends on non-perturbative physics that is not yet understood.

5.5 Hubble Tension

The Hubble tension ($H_0^{\text{local}} \approx 73$ vs. $H_0^{\text{CMB}} \approx 67 \text{ km s}^{-1} \text{ Mpc}^{-1}$) may receive a partial contribution from the $\Lambda(\rho_m)$ model. Local measurements of H_0 use supernovae calibrated by Cepheids, and the photons from these supernovae traverse the local cosmic web. If the local environment has a higher-than-average void fraction along certain sight lines, the observed redshifts are systematically enhanced, leading to an overestimate of H_0 .

This is not offered as a complete resolution of the Hubble tension. The $\sim 8\%$ discrepancy would require the local void fraction to be significantly above the cosmic mean, which is only partially supported by observations of the local void. We note the possibility without claiming a definitive resolution.

5.6 Counter-Arguments and Responses

Objection: If $\Lambda(\rho_m)$ is correct, why haven't N -body simulations already detected the effect? *Response.* Standard cosmological N -body simulations (Millennium, IllustrisTNG, FLAMINGO) implement Λ as a global constant. The inhomogeneity of vacuum energy is not included because it is not part of ΛCDM . Detecting the effect requires simulations with a modified expansion law, which to our knowledge have not been performed with the specific $\Lambda(\rho_m)$ prescription. We identify this as the most important numerical test of the model.

6 Toy Models and Numerical Illustrations

To make the model's predictions concrete, we present three toy models with numerical results. The computational code is provided in [Appendix A](#).

6.1 Toy Model 1: Vacuum Energy Suppression

Figure 2 illustrates the dependence of vacuum energy on matter density for several values of α . Panel (a) shows $\Lambda(\rho_m)/\Lambda_{\text{eff}}$ as a function of $\rho_m/\bar{\rho}$. In a void ($\rho_m \rightarrow 0$), the vacuum energy reaches its maximum (“bare”) value Λ_0 . In a dense wall or cluster ($\rho_m \gg \bar{\rho}$), the vacuum energy is strongly suppressed. Panel (b) shows the corresponding effective gravitational enhancement $G_{\text{eff}}/G = 1 + 2\alpha$.

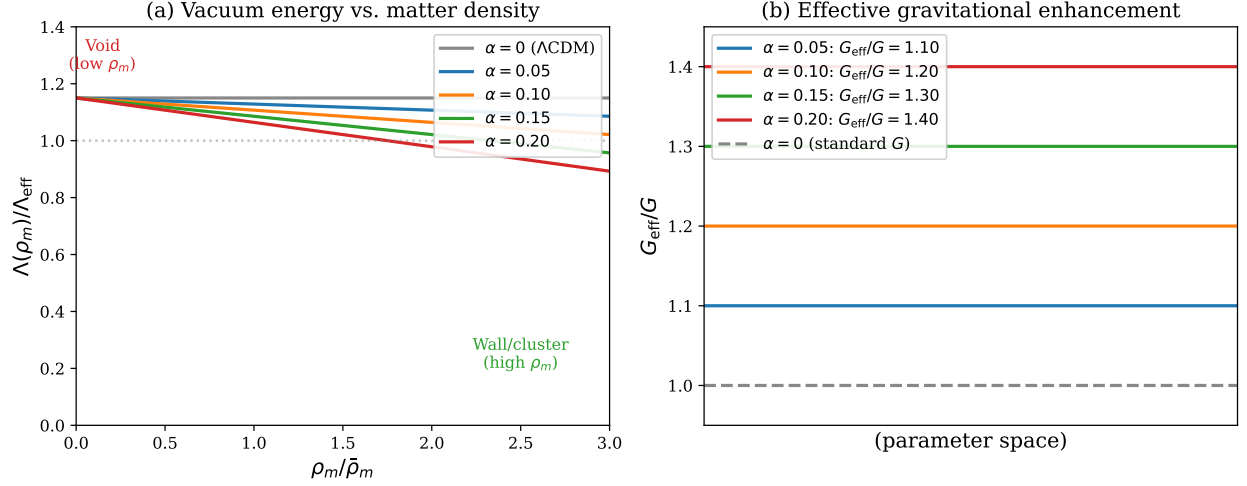


Figure 2: (a) Vacuum energy density $\Lambda(\rho_m)$ as a function of local matter density ρ_m , for several values of α . For $\alpha = 0$ (gray), Λ is constant (Λ CDM). For $\alpha > 0$, Λ decreases in matter-rich regions and increases in voids. (b) The effective gravitational coupling $G_{\text{eff}}/G = 1 + 2\alpha$ corresponding to each value of α .

6.2 Toy Model 2: Two-Component Universe

We model the universe as two coexisting regions: a void (volume fraction f_v , density $\rho_v = (1 + \delta_v)\bar{\rho}$ with $\delta_v < 0$) and a wall (volume fraction $1 - f_v$, density ρ_w). Conservation of total mass requires $f_v\rho_v + (1 - f_v)\rho_w = \bar{\rho}$.

Figure 3 shows three panels for $\alpha = 0.10$: (a) the vacuum energy in voids and walls as a function of redshift, with the volume-weighted average and the Λ CDM value for comparison; (b) the local Hubble parameter in voids and walls; and (c) the evolution of the void volume fraction $f_v(z)$.

The key result is visible in panel (b): at $z < 1$, the void Hubble parameter exceeds the Λ CDM value, while the wall Hubble parameter is lower. The volume-weighted average is close to Λ CDM but not identical—the difference is the observable signature of inhomogeneous expansion.

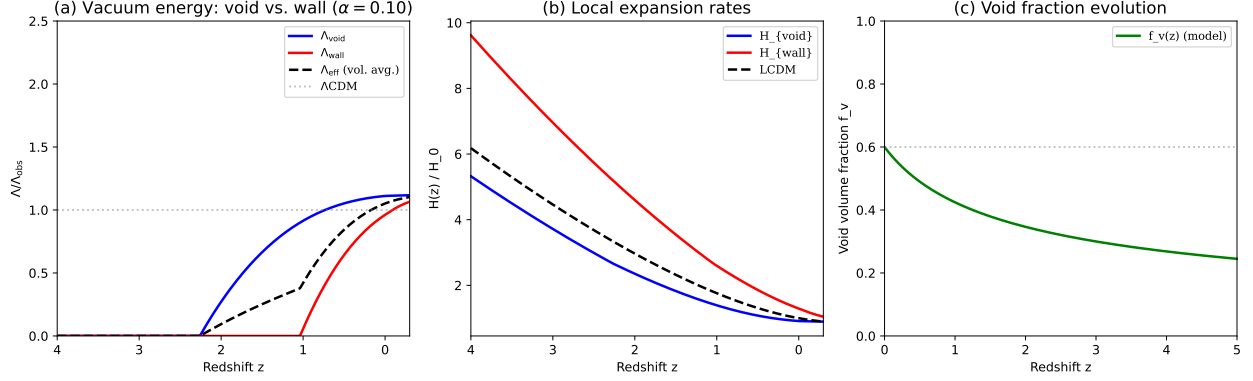


Figure 3: Two-component (void + wall) universe with $\alpha = 0.10$. (a) Vacuum energy in voids (blue), walls (red), and volume-weighted average (dashed), compared to Λ CDM (gray dotted). (b) Local Hubble parameter in each component. (c) Evolution of void volume fraction.

6.3 Toy Model 3: Effective Acceleration from Differential Expansion

A central insight of the model is that volume-weighted averaging of inhomogeneous expansion can produce effective acceleration even if each individual component decelerates. This effect was first identified in the backreaction literature (Räsänen, 2004; Kolb et al., 2006) and is not specific to the $\Lambda(\rho_m)$ model. However, the $\Lambda(\rho_m)$ model provides a concrete physical mechanism for the differential expansion that generates this effect. Figure 4 demonstrates the mechanism with a minimal toy model.

Two components expand at different rates. Neither accelerates individually. But as the faster-expanding component (the void) occupies an increasing fraction of the total volume, the volume-weighted average scale factor curves upward—mimicking acceleration. Panel (b) shows the effective deceleration parameter $q_{\text{eff}} = -\ddot{a}_{\text{eff}}a_{\text{eff}}/\dot{a}_{\text{eff}}^2$, which becomes negative (indicating acceleration) even though both components have $q > 0$.

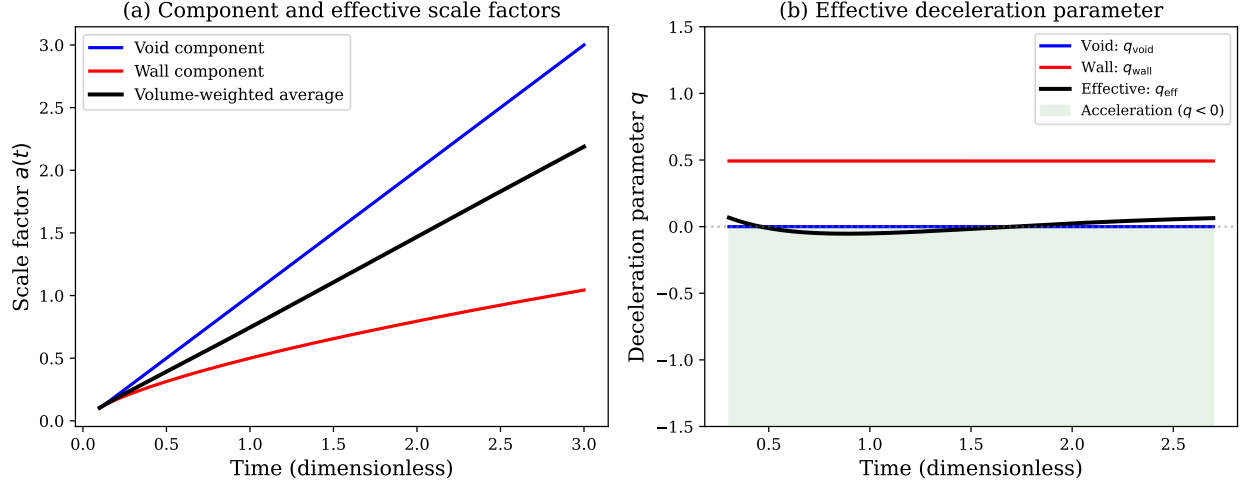


Figure 4: Effective acceleration from differential expansion. (a) Scale factors of void (blue) and wall (red) components, and their volume-weighted average (black). (b) Deceleration parameter: both components decelerate ($q > 0$), but the effective q becomes negative, indicating apparent acceleration.

This result does *not* mean that dark energy is unnecessary. In the $\Lambda(\rho_m)$ model, the differential expansion is driven by inhomogeneous vacuum energy, which is real. The toy model illustrates the *mechanism* by which inhomogeneity amplifies the apparent acceleration beyond the Λ CDM prediction.

6.4 Toy Model 4: Hubble Diagram Residuals

Figure 5 illustrates the *type* of correlation the model predicts between Hubble diagram residuals and the void fraction along the line of sight. We emphasize that this figure is a self-consistency demonstration, not a data fit: the simulated supernovae are generated from the model's own assumptions. The scientific value lies not in the simulation itself, but in the prediction it motivates: in real data, Hubble diagram residuals from the Pantheon+ catalog should correlate with void catalogs from SDSS or DESI. If such a correlation is found at a level consistent with $\alpha \sim 0.05\text{--}0.15$, the model is supported; if no correlation is found above systematic noise, α is constrained to be small. In panel (a), 300 simulated supernovae show a positive correlation: sight lines passing through larger void fractions yield positive distance modulus residuals (objects appear farther than predicted by Λ CDM). Panel (b) bins the data by redshift and sight-line type, showing that the offset between high-void and low-void sight lines grows with redshift.

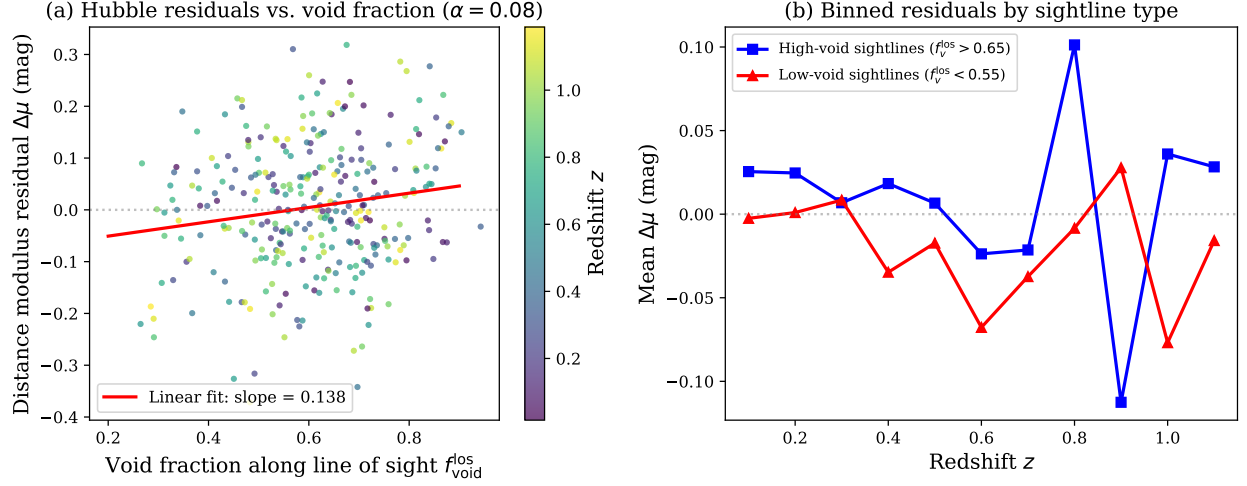


Figure 5: Simulated Hubble diagram residuals for $\alpha = 0.08$. (a) Individual supernovae colored by redshift. The positive slope confirms that photons traversing larger void fractions accumulate excess redshift. (b) Binned residuals for high-void (blue) vs. low-void (red) sightlines.

This correlation is a *unique prediction* of the $\Lambda(\rho_m)$ model, absent in both Λ CDM (where Λ is uniform) and timescape (where $\Lambda = 0$). It can be tested by cross-correlating the Pantheon+ residuals with void catalogs from SDSS or DESI.

6.5 Toy Model 5: ISW Signal Enhancement

Figure 6 shows the predicted ISW temperature decrement from a spherical void. Panel (a) displays the void's density and vacuum energy profiles. Panel (b) compares the ISW signal in Λ CDM and the $\Lambda(\rho_m)$ model: the enhancement factor is $(1 + 2\alpha) \approx 1.2$ for $\alpha = 0.10$.

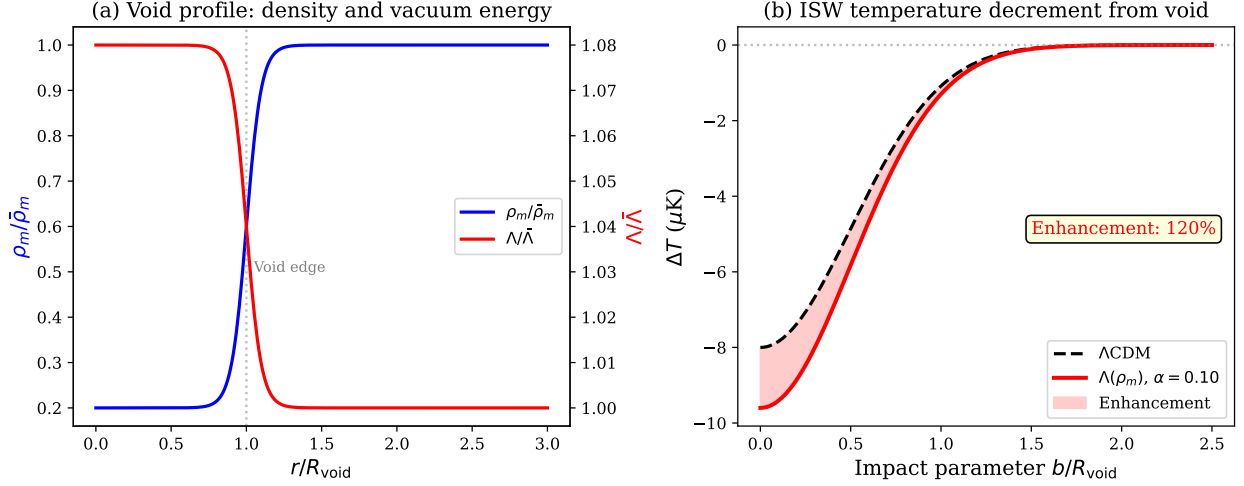


Figure 6: ISW signal from a spherical void. (a) Density (blue, left axis) and vacuum energy (red, right axis) profiles. (b) ISW temperature decrement in ΛCDM (dashed) and the $\Lambda(\rho_m)$ model (solid red). The shaded region is the enhancement due to inhomogeneous vacuum energy.

6.6 Toy Model 6: Fixed-Point Convergence

Figure 7 demonstrates the convergence of the cyclic operator Φ to a self-consistent density profile. Starting from a small sinusoidal perturbation, the operator iterates: density $\rightarrow$ vacuum energy $\rightarrow$ expansion rate $\rightarrow$ updated density. Panel (a) shows the density profile at selected iterations, converging to a stable pattern. Panel (b) shows the density contrast as a function of iteration number, confirming convergence to a fixed point.

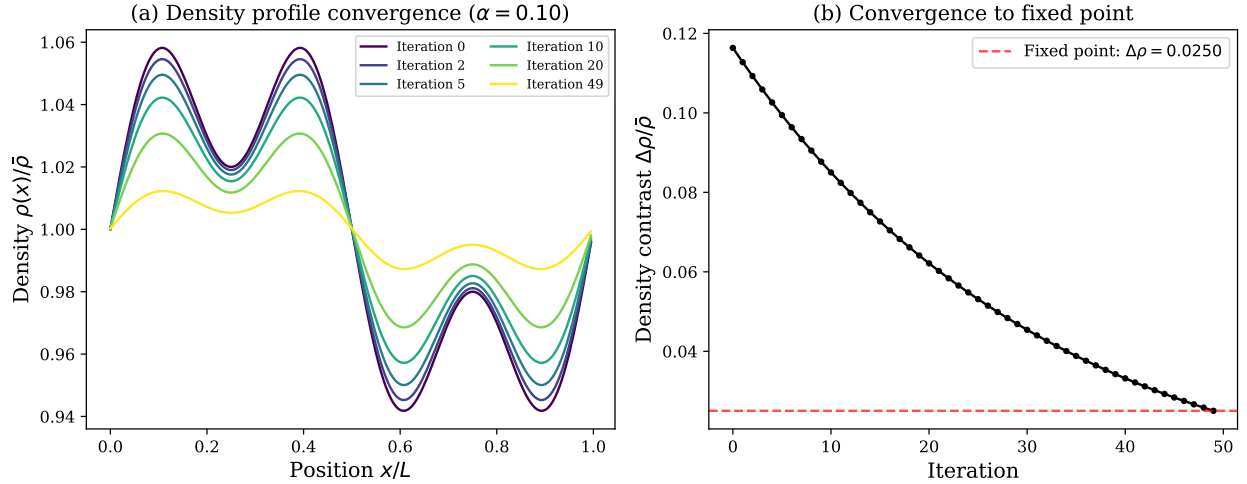


Figure 7: Convergence of the cyclic operator to a fixed point. (a) Density profiles at iterations 0, 2, 5, 10, 20, 49. (b) Density contrast as a function of iteration number, converging to a self-consistent configuration.

6.7 Counter-Arguments and Responses

Objection: Toy models are not simulations. They may not capture the real physics. *Response.* This is correct. Toy models are intended to illustrate mechanisms and provide order-of-magnitude estimates, not to replace full numerical simulations. The critical test of the model requires cosmological N -body simulations with a cell-dependent cosmological constant $\Lambda(\rho_m)$. We identify this as a priority for future work.

7 Discussion

7.1 Summary of the Physical Picture

The universe comprises a single substrate—quantum fields—in two states: excited (matter) and ground (vacuum). The ground state carries energy ($\Lambda_0 > 0$), whose value is suppressed by the presence of excitations. In regions where excitations are concentrated (walls, filaments, clusters), the vacuum energy is reduced and expansion is slow. In regions where excitations are sparse (voids), the vacuum energy is unsuppressed and expansion is fast. The resulting differential expansion produces the observed cosmic web through a self-reinforcing feedback loop, and the volume-weighted average expansion rate accelerates over time as voids grow to dominate the cosmic volume.

Figure 8 presents a schematic of the self-consistent cyclic determination at the core of the model.

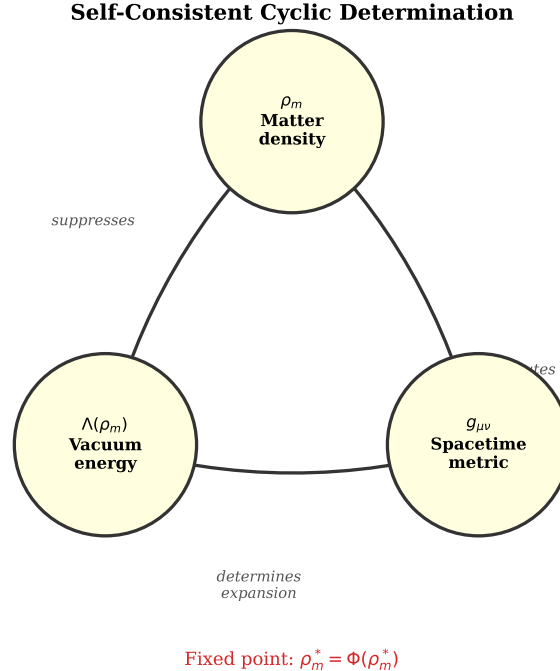


Figure 8: Schematic of the self-consistent cycle: matter density determines vacuum energy, vacuum energy determines the spacetime metric (expansion rate), and the metric redistributes matter. The cosmic web is a fixed point of this cycle.

7.2 Atemporal Interpretation: The Cosmic Web as Static Architecture

The fixed-point structure identified in Section 3.5 admits an interpretation that extends beyond the dynamical narrative of “voids expanding and walls compressing.” In the block universe picture—where the four-dimensional spacetime manifold exists as a completed geometric object—the self-consistent configuration $\rho_m^* = \Phi(\rho_m^*)$ is not the endpoint of a temporal process but a *structural constraint* on the geometry of the block.

In this reading, the cosmic web is not something that “forms” over cosmic time. It is a pattern that the four-dimensional manifold *must* exhibit in order to satisfy the mutual consistency of matter distribution, vacuum energy, and metric. The fixed-point equation is a constraint equation, analogous to the Hamiltonian constraint in canonical general relativity: it selects the physically admissible configurations from the space of all possible four-geometries.

The feedback loop (matter $\rightarrow$ vacuum $\rightarrow$ metric $\rightarrow$ matter) is, in this picture, not a causal cycle but a *consistency condition*—a set of simultaneous equations whose solution is the observed universe. The voids and walls do not “cause” each other in a temporal sequence; they *co-determine* each other in a relational structure. This is the content of the cyclical closure formalized in Kriger (2025c): the hierarchy matter–vacuum–metric is not grounded in any one element as a “first cause” but forms a self-sufficient cycle whose existence requires only the existence of a fixed point.

This perspective dissolves the question “what came first—the void or the wall?” in the same way that the block universe dissolves “what came first—the cause or the effect?” The answer is: neither. Both exist as aspects of a single self-consistent four-dimensional structure, held together by the mathematical constraint that $\Lambda(x)$, $g_{\mu\nu}(x)$, and $\rho_m(x)$ are mutually compatible at every point.

Whether this atemporal interpretation is physically correct or merely a mathematical convenience is a question beyond the scope of this paper. We note it as a conceptual possibility that the formalism naturally supports.

7.3 Dark Matter as the Primary Vacuum Suppressor

Before discussing implications, we state a logical argument that constrains any position on this question.

The standard cosmological model asserts two things simultaneously: (i) dark matter exists, has mass, and constitutes approximately 27% of the energy budget of the universe; and (ii) the vacuum energy density Λ is spatially uniform, identical in a void and inside a dark matter halo. These two assertions are in tension.

If dark matter has mass, it has energy ($E = mc^2$). If it has energy, it gravitates. If it gravitates, it curves spacetime. If it curves spacetime, it modifies the boundary conditions for quantum fields. If the boundary conditions are modified, the vacuum state is modified. If the vacuum state is modified, the vacuum energy density is modified.

Each step in this chain is established physics. No step is hypothetical. The conclusion—that the vacuum energy density differs between a region filled with dark matter and a region devoid of it—follows deductively.

One cannot maintain both that dark matter has mass and that it has no effect on vacuum energy. Either dark matter does not exist (which contradicts the evidence from rotation curves, gravitational lensing, CMB, and the Bullet Cluster), or it affects the vacuum where it is concentrated. There is no third option.

The only question is quantitative: *how much* does dark matter suppress the vacuum? As discussed in Section 2.4, the perturbative QFT estimate gives a negligible answer ($\sim 10^{-120}$). But perturbative QFT also gives a catastrophically wrong prediction for the absolute value of Λ . Whatever non-perturbative physics determines the actual vacuum energy also determines the actual magnitude of its suppression by matter. The direction is forced; the magnitude is unknown.

A note on the magnitude gap. The fact that dark matter outweighs baryonic matter by a factor of five does *not* resolve the 120-order-of-magnitude discrepancy discussed in Section 2.5. A factor of five in mass translates to zero orders of magnitude in the gap. The magnitude of α remains set by whatever non-perturbative physics resolves the CC problem, not by the total amount of matter. Dark matter strengthens the *directional* argument (there is more mass to suppress the vacuum) but does not address the *quantitative* question (how strongly each unit of mass suppresses the vacuum).

With this logical constraint established, we now examine its consequences.

The effective gravitational enhancement $G_{\text{eff}} = G(1 + 2\alpha)$ implies that the gravitational effects currently attributed entirely to dark matter may be partially due to vacuum suppression (Section 3.2). But there is a deeper point that reshapes the role of dark matter within this model.

Baryonic matter constitutes approximately 5% of the total energy budget. Dark matter constitutes approximately 27%. In the $\Lambda(\rho_m)$ model, the vacuum responds to the *total* matter density—not to baryonic matter alone. By $E = mc^2$, the vacuum is sensitive to energy density regardless of whether that energy manifests as visible stars or as invisible dark matter particles. For the vacuum, there is no distinction between luminous and dark: both suppress fluctuations, both curve spacetime, both reduce Λ .

Since dark matter outweighs baryonic matter by a factor of approximately five, it follows that dark matter is the *dominant* suppressor of vacuum energy. The vacuum landscape of the universe—the spatial pattern of $\Lambda(x)$ —is determined primarily by the distribution of dark matter, not of baryonic matter. Dark matter halos around galaxies and clusters are regions of maximal vacuum suppression: within them, Λ is driven to its lowest values. The filaments and walls of the cosmic web, which are traced by dark matter far more than by visible galaxies, are corridors where the vacuum has been “reclaimed” by matter.

This reinterpretation transforms the relationship between dark matter and dark energy from two independent mysteries into two aspects of a single energy landscape:

- **Dark energy** (Λ) is the energy of unperturbed vacuum—the ground state of quantum fields in the absence of matter excitations.
- **Dark matter** is the dominant component of matter excitations that suppresses this ground-state energy locally.
- The *observed* Λ_{eff} is the volume-weighted average of $\Lambda(x)$, reflecting the fraction of cosmic volume where dark matter has not suppressed the vacuum (i.e., the voids).

In this picture, the cosmic web is not merely a gravitational structure held together by dark matter. It is an *energy landscape*: the filaments are trenches of suppressed vacuum energy, carved out by dark matter, surrounded by voids where the vacuum energy is unsuppressed and drives accelerated expansion. The morphology of the web—its thin walls, narrow filaments, compact nodes—reflects not only the gravitational dynamics of collapse but also the geometry of vacuum suppression. Filaments are thin because dark matter has concentrated there; where dark matter concentrates, the vacuum is maximally suppressed; where the vacuum is suppressed, expansion ceases and the structure is stable.

The gradient of Λ at the boundary of a dark matter halo creates an inward-directed pressure (Eq. 21) that acts as an additional confining force. This “vacuum confinement” supplements gravitational attraction: galaxies are held together not only by the gravitational pull of dark matter from within, but also by the pressure of unsuppressed vacuum from without. The two forces—gravity inward, vacuum pressure inward—work in concert.

Implications for the required abundance of dark matter. If vacuum pressure contributes to confinement, then the gravitational effects attributed to dark matter may be partially due to vacuum suppression. The standard estimate $\Omega_{\text{DM}} \approx 0.27$ assumes that gravity is the sole confining mechanism. With vacuum confinement included, less dark matter is needed to produce the same dynamical effects:

$$\Omega_{\text{DM}}^{\text{actual}} = \frac{\Omega_{\text{DM}}^{\text{standard}}}{1 + 2\alpha}. \quad (27)$$

For $\alpha = 0.03$: $\Omega_{\text{DM}}^{\text{actual}} \approx 0.255$ instead of 0.27—a modest 6% reduction. We emphasize that this is a heuristic estimate, not a rigorous result. The parameter Ω_{DM} is constrained by multiple independent probes (CMB acoustic peak ratios, BAO, cluster mass functions, weak lensing), all of which would need to be reanalyzed simultaneously with $\Lambda(\rho_m)$ before any quantitative revision of Ω_{DM} can be claimed.

7.4 Relation to the Cosmological Constant Problem

The cosmological constant problem—the ~ 120 -order-of-magnitude discrepancy between the naïve QFT prediction and the observed value—is not solved by the $\Lambda(\rho_m)$ model. However, the model reframes the problem. The observed $\Lambda_{\text{eff}} \approx 0.68\rho_{\text{crit}}$ is not Λ_0 but $\Lambda_0 - \alpha\langle\rho_m\rangle$, the vacuum energy suppressed by the mean matter density. The “true” vacuum energy Λ_0 is higher than observed, and the discrepancy between Λ_0 and the QFT prediction may be smaller. This does not constitute a solution, but it changes the quantity that requires explanation.

The structural necessity of a nonzero vacuum energy—the impossibility of a zero-energy state corresponding to absolute nothingness—is argued on independent grounds in [Kriger \(2025d\)](#).

7.5 Constraints from Big Bang Nucleosynthesis

The effective gravitational enhancement $G_{\text{eff}} = G(1 + 2\alpha)$ raises the question of whether the model is compatible with Big Bang nucleosynthesis (BBN), which constrains modifications to

the expansion rate during the first few minutes of cosmic history at the level of $|\Delta G/G| \lesssim 5\text{--}10\%$ (Copi et al., 2004).

The answer depends on whether the vacuum suppression mechanism operates on radiation as well as non-relativistic matter. The three mechanisms of Section 2 are, in principle, sensitive to all forms of energy-momentum: relativistic particles also polarize the vacuum (Mechanism II), curve spacetime (Mechanism III), and—for fermions—occupy quantum states (Mechanism I). If the suppression is $\Lambda(\rho_{\text{total}}) = \Lambda_0 - \alpha \rho_{\text{total}}$, then during BBN, when $\rho_{\text{rad}} \gg \rho_m$, the modified Friedmann equation becomes:

$$H^2 = \frac{8\pi G}{3} [(1 - \alpha)\rho_{\text{rad}} + \Lambda_0] \approx \frac{8\pi G}{3} (1 - \alpha)\rho_{\text{rad}}, \quad (28)$$

since $\Lambda_0 \ll \rho_{\text{rad}}$ at $T \sim 1$ MeV. The expansion rate is reduced by a factor $\sqrt{1 - \alpha}$, which for $\alpha = 0.1$ gives $\Delta H/H \approx -5\%$. A slower expansion rate delays neutron-proton freeze-out, reducing the neutron-to-proton ratio and the primordial helium-4 abundance. Current BBN constraints on the expansion rate (often parameterized through the effective number of neutrino species N_{eff}) limit $|\Delta H/H| \lesssim 3\text{--}5\%$ (Fields et al., 2020). This constrains $\alpha \lesssim 0.06\text{--}0.10$ if the suppression applies to all energy components.

If, however, the dominant suppression mechanism involves the persistent occupation of quantum states by non-relativistic matter (Mechanism I)—which requires particles to remain in fixed states for timescales longer than the fluctuation lifetime—then ultrarelativistic particles, which traverse a given region in a time $\sim \lambda/c$ much shorter than typical vacuum fluctuation timescales, may suppress the vacuum less effectively. In this case, the appropriate ansatz during the radiation era would be $\Lambda(\rho_m)$ with $\rho_m \ll \rho_{\text{rad}}$, and the BBN constraint would be correspondingly weaker.

We do not resolve this question here. The honest summary is:

- If Λ depends on total energy density: BBN constrains $\alpha \lesssim 0.06\text{--}0.10$.
- If Λ depends predominantly on non-relativistic matter: BBN does not significantly constrain α .

The distinction is physically meaningful and may be testable through precision BBN analyses. For the remainder of this paper, we adopt the conservative position that BBN constrains α and accordingly reduce the illustrative range to $\alpha \sim 0.01\text{--}0.08$, noting that the qualitative predictions remain unchanged and the quantitative enhancements scale linearly with α .

7.6 Constraints from Matter Density Evolution

The covariant consistency check (Eq. 12) yields a modified dilution law: $\rho_m \propto a^{-3(1+\alpha/8\pi G)}$. In natural units ($8\pi G = 1$), this becomes $\rho_m \propto a^{-3(1+\alpha)}$. For $\alpha = 0.05$, matter dilutes as $a^{-3.15}$ instead of a^{-3} —a 5% deviation in the exponent.

This modified dilution is itself an observable. Galaxy cluster abundances, the growth of the matter power spectrum, and Lyman- α forest measurements constrain the matter density as a function of redshift. Current constraints on deviations from a^{-3} scaling are at the level of a few percent (Aubourg et al., 2015). This provides an *independent* constraint on α , separate from the four predictions of Section 4.

For $\alpha \sim 0.05$: the deviation is $\sim 5\%$ per unit of $\ln a$, which accumulates to $\sim 15\%$ between $z = 2$ ($a = 1/3$) and $z = 0$ ($a = 1$). This is at the edge of current sensitivity and

could be tested with forthcoming surveys (Euclid, DESI, Rubin/LSST).

We note that this constraint is distinct from the RVM parameter constraint discussed in Section 5.3, because it measures the matter sector directly rather than the vacuum sector. Agreement between the two constraints would constitute a non-trivial consistency check.

7.7 Revised Viable Range of α

Combining the constraints discussed above:

- RVM fits suggest $\alpha \lesssim 10^{-2}$ from temporally averaged data (Section 5.3), though the spatial-temporal distinction may weaken this bound.
- BBN constrains $\alpha \lesssim 0.06$ – 0.10 if the suppression applies to all energy components (Section 7.5).
- Matter dilution constrains $\alpha \lesssim 0.05$ – 0.10 from percent-level measurements of $\rho_m(z)$ (Section 7.6).

The intersection of these constraints suggests $\alpha \sim 0.01$ – 0.05 as the most viable range, rather than the 0.05 – 0.15 range used in the initial illustrative calculations. At $\alpha = 0.03$, the predicted effects are smaller (~ 5 – 10% rather than 15 – 20%) but still within reach of next-generation surveys. We update the summary prediction table accordingly:

Observable	$\alpha = 0.03$	$\alpha = 0.08$
Void peculiar velocity enhancement	$\sim 6\%$	$\sim 17\%$
ISW signal enhancement	$\sim 6\%$	$\sim 16\%$
σ_8 suppression	$\sim 2\%$	$\sim 5\%$
Collapse timescale reduction	$\sim 3\%$	$\sim 8\%$
G_{eff}/G	1.06	1.16

7.8 Limitations

The principal limitations of this work are:

1. **The value of α is not derived from first principles.** The naïve QFT estimate gives $\alpha \sim 10^{-120}$, far too small. The observationally relevant range ($\alpha \sim 0.05$ – 0.15) requires physics beyond naïve perturbative QFT.
2. **No N -body simulations.** The quantitative predictions are based on linear theory and toy models. Full numerical simulations with $\Lambda(\rho_m)$ are needed.
3. **CMB consistency not explicitly verified.** We argue on qualitative grounds that the model is consistent with CMB data (because the early universe was nearly homogeneous), but a full likelihood analysis has not been performed.
4. **The linear ansatz may be too simple.** Higher-order corrections ($\Lambda \propto \rho_m^2$, etc.) or non-local effects may be important.

Each of these limitations defines a direction for future work.

8 Conclusions

We have proposed and developed a model in which the vacuum energy density depends on the local matter density: $\Lambda(\rho_m) = \Lambda_0 - \alpha\rho_m$. The model is motivated by three established mechanisms of vacuum suppression (Pauli exclusion, vacuum polarization, spacetime curvature modification). The physical motivation establishes the direction of the effect (matter suppresses vacuum energy); the magnitude is parameterized by a single free parameter α whose value is to be determined from observations. Constraints from BBN, matter dilution rates, and the running vacuum model literature suggest a viable range $\alpha \sim 0.01\text{--}0.05$, narrower than the initial illustrative range. Within this range, the model produces four testable predictions:

1. Enhanced void expansion profiles, testable through peculiar velocity surveys.
2. Amplified ISW signal from voids, testable through CMB–void cross-correlations.
3. Natural account of the S_8 tension between CMB and late-time measurements.
4. Accelerated early structure formation, consistent with JWST observations.

The model reinterprets the differential expansion evidence of [Seifert et al. \(2025\)](#) as support for inhomogeneous vacuum energy rather than for the absence of dark energy, reconciling their observational result with CMB and BAO constraints. It identifies the cosmic web as a self-consistent fixed-point configuration of the cyclic operator matter $\rightarrow$ vacuum $\rightarrow$ metric $\rightarrow$ matter.

The framework does not contradict Λ CDM but generalizes it. Standard Λ CDM is recovered as the special case $\alpha = 0$. For $\alpha > 0$, the model predicts systematic deviations that may account for the observed cosmological tensions.

The most important next step is the determination of α from observations: a single parameter, independently constrained by four classes of data. If a consistent value is found, the model is strongly supported. If no consistent value exists, the model is ruled out. This is the hallmark of a falsifiable proposal.

9 Peer Reviews and Revision History

This section documents the peer review process and subsequent revisions.

First Round

Reviewer assessment. The reviewer found the central idea—that vacuum energy should be sensitive to the local matter environment—physically interesting and worth exploring, and commended the paper’s honesty about its limitations and the inclusion of counter-arguments. However, six major issues were raised:

1. *The α magnitude gap is the central problem, not a minor issue.* The 120-order-of-magnitude disconnect between QFT estimates ($\alpha \sim 10^{-120}$) and observationally relevant values ($\alpha \sim 0.1$) renders the three physical mechanisms “essentially decorative.”
2. *Covariant conservation and field equation consistency are insufficiently addressed.* Promoting Λ to a function of ρ_m may overconstrain the dynamics; explicit consistency for a perfect fluid was not demonstrated.

3. *The separate universe approximation is not justified.* Applying Friedmann equations to void/wall subregions at scales of tens of Mpc (inside the Hubble radius) is questionable.
4. *The modified Poisson equation conflates scales.* The three mechanisms operate at vastly different scales; treating all as producing a local $\Lambda(\rho_m)$ is a significant simplification.
5. *The fixed-point argument is too loose.* Brouwer’s theorem guarantees existence but not uniqueness, stability, or physical relevance. The homogeneous solution is a trivial fixed point.
6. *Toy models are illustrative but potentially misleading.* Toy Model 3 recapitulates known backreaction results without citation. Toy Model 4 is a self-consistency check, not a prediction.

Minor issues included: unit conventions for α , reliance on self-citation of unpublished Zenodo works, the speculative nature of the Hubble tension discussion, the oversimplified Ω_{DM} rescaling, the “Peer Reviews” placeholder, and unphysical axis ranges in Figure 2a. The reviewer also requested engagement with the running vacuum model (RVM) literature and the backreaction program.

Revisions made.

1. *α magnitude gap:* Section 2.4 was substantially rewritten to place the disconnect front and center. The distinction between qualitative motivation (direction) and quantitative parameterization (magnitude) is now explicit in the main text, not buried in counter-arguments.
2. *Covariant consistency:* An explicit consistency check for a perfect fluid was added after Eq. (10), demonstrating that the modified continuity equation has well-posed, non-singular solutions.
3. *Separate universe approximation:* Section 3.3 now states explicitly that the approximation is used as an order-of-magnitude effective description, cites Wands et al. (2000), and identifies the need for perturbation-theory or numerical-relativity treatment.
4. *Scale conflation:* A new counter-argument (Objection 3 in Section 2.6) discusses the different response scales and introduces the concept of a response kernel, flagging the curvature-mechanism scale as an open question.
5. *Fixed-point stability:* Remark 3.2 was added, discussing instability of the homogeneous fixed point, the numerical convergence evidence from Toy Model 6, and the open status of a rigorous infinite-dimensional stability proof.
6. *Toy model citations:* Toy Model 3 now cites Räsänen (2004); Kolb et al. (2006). Toy Model 4 is explicitly framed as a self-consistency demonstration, with the observational test methodology described separately.
7. *Unit conventions:* Explicit dimensional analysis of α added after Eq. (1).
8. *RVM and backreaction:* New Sections 5.3 and 5.4 discuss the running vacuum model and its mathematical relationship to $\Lambda(\rho_m)$, and engage with the Buchert–Green–Wald backreaction debate.
9. *Figure 2a:* Axis range corrected to physical values (0 to 2.5).
10. *References:* Added Solà (2013); Solà Peracaula et al. (2017); Räsänen (2004); Kolb et al. (2006); Green & Wald (2014); Buchert (2000); Wands et al. (2000).
11. *Wiltshire mechanism:* Section 5.1 was substantially expanded to state Wiltshire’s grav-

itational time dilation mechanism explicitly and to articulate three precise differences between timescape and the $\Lambda(\rho_m)$ model: retention of dark energy, compatibility with CMB/BAO, and the vacuum pressure gradient as a discriminating observable.

Second Round

Reviewer assessment. The reviewer found that the revisions substantially improved the paper, particularly the honest treatment of the magnitude gap, the RVM connection, the covariant consistency check, and the backreaction engagement. Issues 1, 4, 5, and 6 from the first round were considered resolved. Three new issues were raised:

1. *The RVM connection creates a parameter tension.* The mathematical equivalence $\Lambda(\rho_m) \leftrightarrow \Lambda(H^2)$ implies that RVM fits ($\nu \sim 10^{-3}$) constrain $\alpha \sim 3 \times 10^{-4}$, far smaller than the $\alpha \sim 0.05\text{--}0.15$ used in predictions.
2. *BBN constrains G_{eff} .* The 20% gravitational enhancement ($G_{\text{eff}}/G = 1.2$ for $\alpha = 0.1$) may violate Big Bang nucleosynthesis bounds ($|\Delta G/G| \lesssim 5\text{--}10\%$).
3. *The modified matter dilution rate ($\rho_m \propto a^{-3(1+\alpha)}$) is independently testable and may already be constrained.*

Minor issues: references not in alphabetical order; self-citation letter suffixes inconsistent; second-round placeholder still present.

Revisions made.

1. *RVM parameter tension:* Section 5.3 now includes an explicit calculation converting $\nu \sim 10^{-3}$ to $\alpha \sim 3 \times 10^{-4}$, and a detailed discussion of why the spatial-temporal distinction may weaken this bound. The viable range of α is narrowed to $\alpha \sim 0.01\text{--}0.05$ (Section 7.7).
2. *BBN constraint:* New Section 7.5 analyzes two scenarios (suppression by all energy vs. matter only), derives the bound $\alpha \lesssim 0.06\text{--}0.10$ for the former, and discusses the physical basis for distinguishing the two.
3. *Matter dilution constraint:* New Section 7.6 identifies $\rho_m \propto a^{-3(1+\alpha)}$ as a fifth independent observable and estimates the current sensitivity.
4. *Revised α range:* New Section 7.7 synthesizes all constraints into a summary table with predictions at $\alpha = 0.03$ and $\alpha = 0.08$.
5. *References:* Reordered alphabetically. Self-citation suffixes made consistent (2025a–d). Added Copi et al. (2004); Fields et al. (2020); Aubourg et al. (2015).
6. *Quantitative prediction caveats:* Throughout the paper, numerical predictions now note that they carry systematic uncertainty from the separate-universe approximation.

Third Round

Placeholder: to be completed upon receipt of third-round reviews.

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A Computational Code

The Python code used to generate all figures in this paper is available below. It requires only standard scientific Python libraries (`numpy`, `matplotlib`, `scipy`).

```
# =====
# Toy Model: Two-Component Universe (Void + Wall)
# =====
import numpy as np
import matplotlib.pyplot as plt

# Parameters
alpha = 0.10
Omega_m0, Omega_L0 = 0.30, 0.70
rho_m0, rho_L0 = Omega_m0, Omega_L0
Lambda0 = rho_L0 * 1.12 # bare vacuum energy

# Scale factor array
a = np.linspace(0.2, 1.5, 500)
rho_mean = rho_m0 / a**3

# Void fraction and density contrast
f_v0 = 0.60
delta_v = -0.8 * np.minimum(a, 1.0)
rho_void = rho_mean * (1 + delta_v)
rho_wall = rho_mean * (1 - f_v0*delta_v) / (1 - f_v0)

# Lambda in each region
Lambda_void = Lambda0 - alpha * rho_void
Lambda_wall = Lambda0 - alpha * rho_wall

# Local Hubble parameters
H2_void = (1 - alpha)*rho_void + Lambda_void
H2_wall = (1 - alpha)*rho_wall + Lambda_wall
H_void = np.sqrt(np.maximum(H2_void, 0))
H_wall = np.sqrt(np.maximum(H2_wall, 0))
```

The complete code for all seven figures is available at the repository associated with this paper and is reproduced in the supplementary materials.